Breath of Life: A Biofeedback Game with Diverse Breathing Techniques for Enhanced Real-Life Emotion Regulation

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Abstract

Mental health is a critical public health concern, with emotion regulation (ER) playing a pivotal role in maintaining and improving psychological well-being. While respiratory biofeedback games have emerged as effective tools for ER training, existing research predominantly focuses on single-session interventions with limited techniques, emphasizing short-term skill transfer in simulated environments. This approach has led to a significant gap in understanding the long-term application of these techniques in real-life contexts, which is essential for designing effective ER training games. To address this gap, we developed a biofeedback game incorporating four distinct breathing techniques and conducted a 10-week longitudinal study with 25 participants. After one week of training, the majority of participants demonstrated successful transfer of breathing techniques to real-life scenarios. Our findings provide actionable design insights for promoting long-term skill transfer and highlight the potential of respiratory biofeedback games as sustainable tools for emotion regulation training.

CCS Concepts

• **Human-centered computing → Collaborative and social computing devices**; **Empirical studies in collaborative and social computing**; **Empirical studies in HCI**.

Keywords

Breath, Self-Expression, Breath Sharing, Affective Communication, Social Wearable, Pneumatic Display, Collocated Interaction

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CHI EA '25, Yokohama, Japan
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ACM ISBN 979-8-4007-1395-8/25/04
https://doi.org/10.1145/3706599.3720103

ACM Reference Format:
Qi Chen, Jiachen Du, Zhihao Yao, Yuanling Feng, Guanhong Liu, Ning Ni, Zhenyu Wu, Haoran Liu, Xijing Chen, and Haipeng Mi. 2025. Breath of Life: A Biofeedback Game with Diverse Breathing Techniques for Enhanced Real-Life Emotion Regulation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3706599.3720103

Figure 1: We developed a biofeedback game that utilizes breathing as an input, incorporating four core breathing techniques. A longitudinal study spanning 10 weeks was conducted. The right panel demonstrates how real-time feedback is provided to players, enabling them to dynamically respond to their breathing patterns through game avatars and graphical visualizations.

1 INTRODUCTION

Mental health has emerged as a critical global concern[26], with emotion regulation (ER) recognized as a fundamental skill that significantly impacts individuals' quality of life and overall well-being[8, 11, 19]. Within the HCI community, researchers have explored various approaches to ER training, including biofeedback-based interventions, cognitive-behavioral therapy (CBT) techniques, reflection support, and reminder systems[43]. Among these approaches, breathing techniques have gained prominence as somatic tools for emotional regulation, particularly through biofeedback systems. These multidimensional strengths collectively explain the effectiveness of breathing techniques: immediate physiological and psychological benefits demonstrated in neurocognitive studies, seamless integration into daily routines, and the technical simplicity enabling broad applicability across diverse populations and contexts[14, 22, 52].

Game-based breathing biofeedback training has emerged as an accessible approach, demonstrating significant efficacy in alleviating negative emotions[34, 44]. However, the long-term effectiveness of such training largely depends on users' ability to transfer learned breathing techniques to real-life contexts [49]. Current research remains predominantly confined to simulated environments for assessing skill transferability[34, 47, 51], while empirical investigations in complex, real-life scenarios are notably scarce.

To address this research gap, we designed and developed a biofeedback game named *Breath of Life* (Figure 1), aiming to facilitate the effective application of breathing techniques for emotion regulation in daily contexts. We conducted a 10-week longitudinal study with 25 participants, employing a mixed-methods approach that combines quantitative measures and qualitative feedback through questionnaires and interviews. This study seeks to investigate the following research questions: **RQ1:** How to enhance the daily transfer and long-term application of breathing techniques? **RQ2:** How do participants adapt breathing techniques in diverse daily contexts after game-based training? This paper presents several key contributions, including:

(1) We systematically identify and validate three key design considerations for facilitating skill transfer in digital emotion regulation training, with particular emphasis on long-term application in real-world contexts.
(2) We developed *Breath of Life*, a biofeedback game integrating four breathing techniques through five training sessions.
(3) We conducted a 10-week longitudinal study with 25 participants. The results validated our training approach and provided insights into transfer patterns of multiple breathing techniques.

2 RELATED WORK

2.1 Research on Breathing Training in the Field of HCI

Breathing, as a body-centered response modulation technique and a crucial emotion regulation strategy, has garnered significant attention in the HCI community[43]. Researchers are committed to enhancing breathing training through technological means such as biofeedback systems, gamified interventions, wearable devices, and haptic interactions[16, 21, 38, 46]. Training users to reduce and control their breathing frequency is one of the key focuses of research in this field [15, 32, 34, 44, 51]. For example, the *aSpire* clip-enabled mobile pneumatic-haptic device uses subtle haptic feedback to assist users in adjusting their breathing frequency during their commute[15]. In terms of learning special breathing techniques [1, 33], the *Breathe and Rescue* game teaches players how to perform the 4-7-8 breathing method through character dialogues and visual feedback, helping them understand and practice this stress-relief technique[37]. To extend breath length[24, 25], Rockstroh et al. conducted diaphragmatic breathing training using virtual reality games and demonstrated that the system improves awareness and perception of breathing[39]. Prior investigators have explored strategies to enhance engagement in breathing training. For instance, the

Focus Cat game employs gamified mechanics where players accumulate points by performing breathing exercises to rescue virtual cats, thereby increasing adherence to training regimens [41].

However, most existing studies focus on short-term, single-session training, with a limited comprehensive evaluation of the long-term benefits of breathing training. Therefore, more extensive longitudinal studies are required to thoroughly understand the actual effects of breathing training in long-term daily applications[43, 50].

2.2 Breathing Biofeedback Games for Promoting the Transfer of Emotional Regulation Abilities

Breathing biofeedback games have proven effective in improving emotional regulation[3, 27, 39, 51], **but the key to realizing the long-term benefits of breathing training lies in transferring these techniques to real-life contexts**[43]. Existing research on biofeedback games for breathing training mainly focuses on skill transfer within simulated environments. For example, Zafar et al. conducted an 18-minute experiment. Their results demonstrated that participants retained breathing control abilities acquired during training, even when executing short-term stress tasks post-intervention[51]; Wang et al. conducted a 60-minute experimental study. The findings demonstrate that participants successfully transferred acquired deep breathing skills to simulated driving scenarios following training[47]; Marieke M. J. W. et al. designed a 4-week VR training protocol for police officers, involving ten structured sessions. Post-experiment analysis revealed successful transfer of slow breathing techniques to non-biofeedback virtual environments, demonstrating skill retention beyond training conditions[34]; Silva et al. conducted a 3-day user study demonstrating that the gamified breath feedback (GBF) group achieved superior skill transfer to post-training contexts, with higher practice frequency and enhanced self-reported relaxation compared to the paced breathing (PACE) control group[42].

While these studies demonstrate the effectiveness of breathing technique transfer in specific simulated scenarios and tests (e.g. the Stroop color-word test and mental arithmetic), there remains a gap in evaluating the long-term impact of game-based training on the application of these techniques in real-life contexts.

3 DESIGN CONSIDERATIONS FOR FACILITATING SKILL TRANSFER IN REAL-LIFE CONTEXTS

In daily life, individuals often develop relatively fixed emotion regulation patterns[23]. Therefore, the transfer of emotion regulation skills can be conceptualized as a dynamic process where individuals actively adopt new behavioral strategies to achieve emotional balance during states of emotional regulation[10]. This process essentially constitutes behavioral change. According to the Fogg behavior model (FBM), behaviors with lower complexity and higher ability are more likely to be triggered[20]. Based on this, we propose three design considerations aimed at enhancing the ability to facilitate skill transfer in real-world contexts.

3.1 Design Consideration 1: Experiencing the Impact of Emotion Regulation through Training

Existing work in the HCI field in digital emotion regulation training includes both didactic and experiential approaches. Biofeedback, as a key component of experiential approaches, primarily focuses on eliciting the user's subconscious emotion regulation[43]. For these techniques to be effective, participants must perceive their impact during training, as this positive experience can significantly enhance the motivation to apply the skills in real-life situations[43].

3.2 Design Consideration 2: Enhancing Ability to Facilitate Skill Transfer in Real-Life

(1)Multiple training sessions: Single-session training often proves insufficient for achieving precise control over breathing rhythm and frequency. Prior research indicates that paced breathing techniques (4–10 breaths per minute) may initially induce discomfort in participants and exhibit limited long-term adherence [40]. However, current research often relies on single-session training to develop breathing techniques [31, 44, 51], which is inadequate for achieving skill mastery and long-term transfer. Consequently, the implementation of repetitive training is essential in our design consideration to facilitate reliable skill transfer. **(2)Progressive sequence from simplicity to complexity:** In the context of the FBM, when users encounter a challenge that matches their increasing ability, their motivation remains high[20]. Conversely, if the complexity of the game challenge increases too quickly without an adequate boost in ability, users may become demotivated. **(3)Balancing emotional dynamics in training game design:** Research indicates that training breathing techniques in stress-inducing game scenarios can more effectively facilitate the transfer of breathing techniques [34, 51]. Excessive stress could cause negative emotional impacts. Therefore, game design should balance relaxation and tension to optimize the training and ensure a positive user experience.

3.3 Design Consideration 3: Multiple Techniques in Game Training for Personalized Real-Life Application

The ER training requires a variety of techniques to address different situational demands and individual needs. Firstly, different breathing techniques are suited to various real-life situations, emotional states, and individual needs[7]. Secondly, research shows that breathing patterns are highly diverse and personalized[6], with individual physiological and psychological differences influencing breathing preferences[30]. However, most recent serious games for breathing training typically offer only 1-2 techniques[1, 13, 24, 28, 33, 34, 46]. To help users experience the benefits of different breathing techniques, games should incorporate a variety of techniques, enabling users to find the methods that best suit their needs and promote skill transfer to real-life contexts[7].

4 BREATH OF LIFE

4.1 Game-Based Biofeedback Training for Breathing Techniques

Our game is compatible with various devices. Respiratory signals are collected using headphones, ensuring accurate detection of exhalation, inhalation, and breath-hold. Building upon our design considerations, the detailed implementation process is systematically outlined in Figure 2. We integrated four validated breathing techniques (Long breathing[50], 4-7-8 breathing technique[2, 35, 36, 48], Tactical breathing[9]and Paced breathing[12, 22, 29, 50]) into a progressive sequence based on their difficulty levels, starting with simpler techniques and gradually advancing to more complex ones. Empirical studies suggest that 5-minute biofeedback sessions practiced at least six times weekly are particularly effective in promoting emotional regulation improvements [4, 7, 18]. Based on this evidence, we designed a training protocol with seven sessions in total, each lasting 5-12 minutes.

4.2 Level Design

We developed the biofeedback game with five levels: **(1)Establishing breathing awareness:** During this phase, users complete the acquisition of biological signals, develop foundational breathing awareness, and familiarize themselves with the core game-play mechanics. **(2)Long-breath training:** Users navigate the game world using keyboard controls and employ breathing techniques to cast magical leaves, healing mythical creatures.**(3)Breathing control under tension:** This stage focuses on training players to maintain slow and steady breathing in high-stress situations, thereby enhancing their ability to regulate emotions under pressure. Players must use controlled, slow breaths to effectively respond to the pursuit of a challenging monster. **(4)Healing mode and advanced breathing techniques:** In Healing mode, players perform specialized breathing exercises for relaxation and recovery. **(5)Forest rescue and paced breathing exercises:** Forest Rescue mode involves specific slow-breathing exercises to restore the balance of the forest.

5 EXPERIMENTAL METHODS

Participants: The study recruited 25 undergraduate students, with an average age of 20 (±2) years. Eligibility criteria required no history of mental illness or self-harm. Both the consent form and the study protocol were approved by the University Committee on Science and Humanities Ethics (Approval No: THU-04-2024072, Approval Date: July 1, 2024).

Measurement methods: To comprehensively capture both short-term behaviors and long-term trends, we combined intensive data collection (1 week of daily self-report questionnaires) with longer-interval data collection (monthly interviews and questionnaires) to avoid further reminder effects. One-month interval balances memory reliability, behavior maintenance, and participant burden and optimizing the assessment of skill transfer and long-term changes[5]. These methods allow for the analysis of dynamic changes in behavior while also tracking sustained effects over time [5, 17, 23, 45]. However, due to the lack of established scales for measuring breathing skill transfer, we employed a custom-designed questionnaire to assess participants'usage of breathing techniques in real-life contexts. The questionnaire consisted of yes/no questions, questions about which breathing techniques were used, open-ended questions, and did not involve any scoring or rating assessments.

The experimental procedure: The study lasted 10 weeks and was divided into four phases (Figure 3).

(1) **Pre-test:** Participants completed the breathing skills and usage questionnaire to assess their baseline knowledge and usage of breathing techniques (see Questionnaire 1 in the Appendix).
(2) **Game training**: During the experiment, participants used computers and headphones provided by the research team to complete game tests. The average gameplay duration for each test was 5 to 20 minutes. Additionally, after the everyday game test, researchers conducted a 5-minute semi-structured interview with the participants (see Interview Outline 1 in the Appendix).
(3) **Post-test (Week 2):** After completing the training, on the morning of the eighth day, participants completed a brief two-question questionnaire to retrospectively assess their frequency of applying breathing techniques learned from the ***"Breath of Life"*** game in their daily lives and to identify the specific techniques they used during the past training week (see Questionnaire 2 in the Appendix). Additionally, participants were asked to complete a daily self-report for one week. To ensure the accuracy of the self-reports, participants were asked to recall their use of breathing techniques each evening. Recognizing that participants had varying schedules, we requested that users submit their digital self-reports each evening between 9:00 pm and 11:00 pm. The self-report included five short questions on whether breathing techniques were used that day, the contexts and techniques applied, the participant's subjective experience, and their motivation for using the techniques (see Questionnaire 3 in the Appendix).
(4) **Long-term follow-up (Week 3-10):** On the final day of the sixth and tenth weeks, participants completed a questionnaire assessing their usage of breathing techniques learned from the game over the past four weeks. The questionnaire included questions on whether they used the techniques (consciously or unconsciously), the frequency of use, specific techniques applied, the contexts of use, and reasons for not using the techniques if applicable (see Questionnaire 4 in the Appendix). Based on their responses, participants were invited to participate in semi-structured interviews. The interviews aimed to explore participants' mastery of breathing techniques, their perceived effectiveness for emotional regulation, differences between in-game and real-life applications, and their continued usage in daily life, including specific contexts and deeper motivations(see Interview Outline 2 in the Appendix).

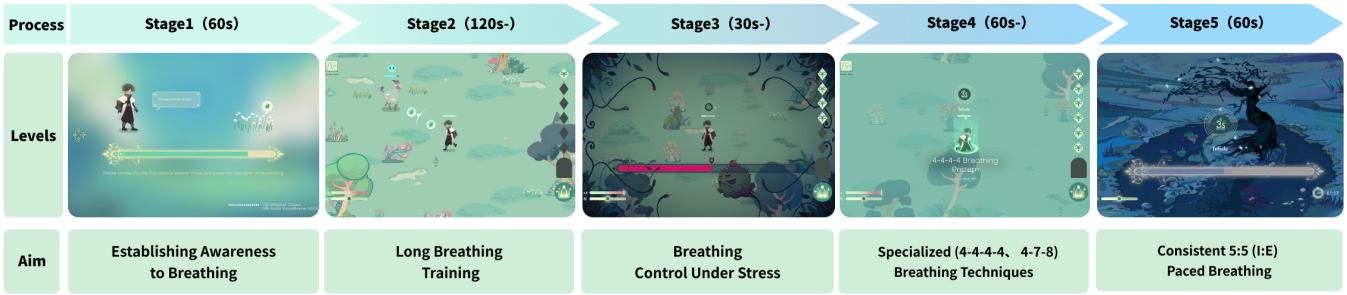

Figure 2: Process of game-based training

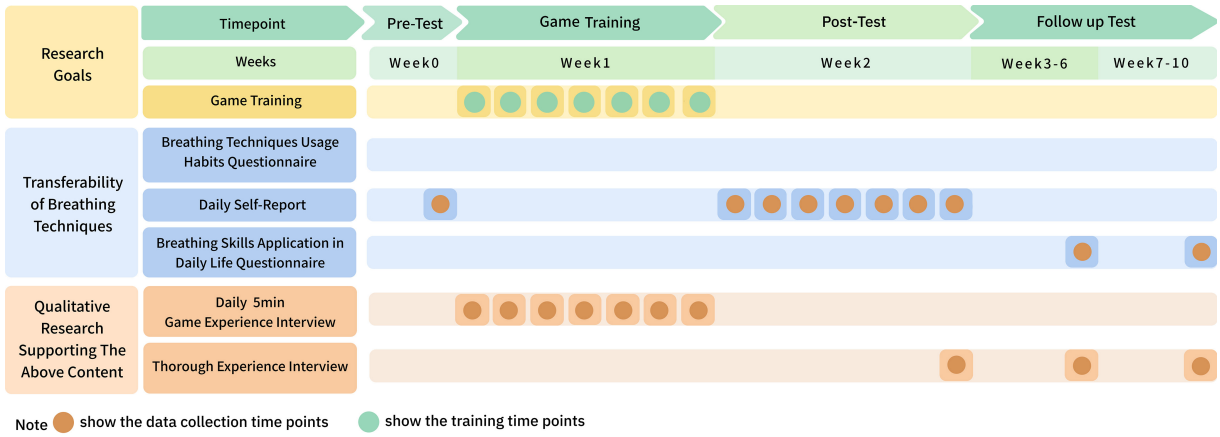

Figure 3: Experimental procedure

6 RESULTS

6.1 Quantitative Results

6.1.1 Number and Frequency of Participants Demonstrating Transfer.

(1) **Proportion of participants who applied breathing techniques in real life:** Most participants successfully transferred and continued to use the techniques in their daily lives (Figure 4(a)). Additionally, a longitudinal comparison of participants showed that 84% of participants (n=21) exhibited technique transfer within 10 weeks.

(2) **Frequency of breathing technique application across time points**: Following game-based training, a gradual decline in breathing technique application frequency was observed. Specifically, during the initial week of training, among the 19 participants who demonstrated technique transfer, 8% (n=2) used the techniques for 4-5 days, 40% (n=10) for 2-3 days, and 28% (n=7) for 1 day. In the week following training, only 1 participant used the techniques daily, 1 participant used them for 6 days, 24% (n=6) for 4-5 days, 32% (n=8) for 2-3 days, and 16% (n=4) for 1 day. By week 3-6, 32% (n=8) of participants were using the techniques regularly each week, 4% (n=1) almost daily, 8% (n=2) 3-5 times per week, and 20% (n=5) 1-2 times per week. Additionally, 24% (n=6) used the techniques sporadically, and 8% (n=2) only in specific situations.

6.1.2 Usage of Various Breathing Techniques During the Transfer Process.

(1) **The proportion of participants using multiple breathing techniques across different time points (week 2 to week 9):** A split is observed between those using multiple techniques and those sticking to one (Figure 4(b))

(2) **Longitudinal comparison of the breathing techniques used by the same participant from 9 weeks**: Among the 21 participants who exhibited transfer, 8 participants used 3-4 techniques, 7 participants used two techniques, and 6 participants consistently used one technique. This highlights the variability in the long-term use of multiple breathing techniques among individuals. Furthermore, the use of breathing techniques by individuals fluctuated over the 9-week period, with participants frequently switching between different techniques.

6.2 *Qualitative Results*

We gathered qualitative feedback from participants through semi-structured interviews to understand how game-based training facilitates the transfer of breathing techniques as emotion regulation

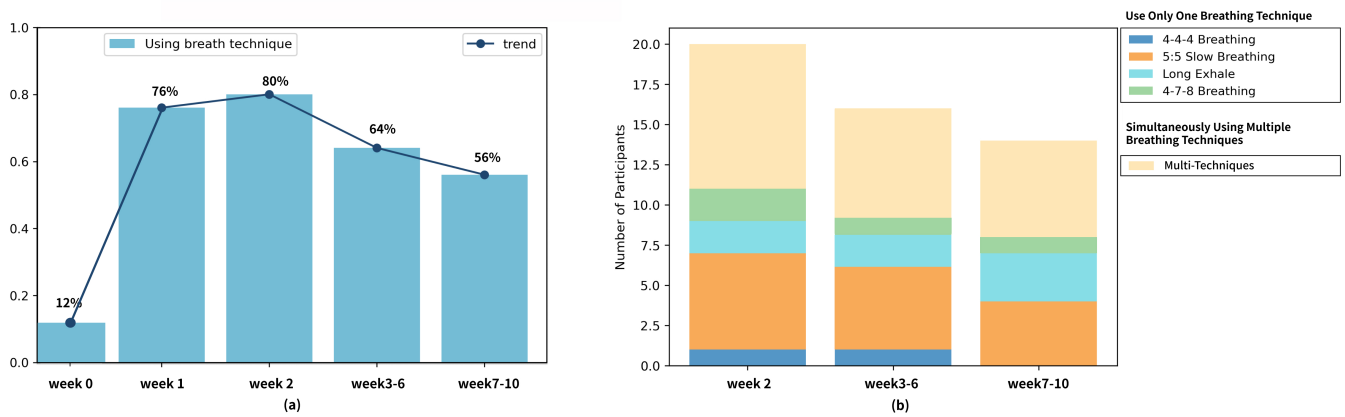

Figure 4: (a) Prior to the experiment, only 12% (n=3) of participants employed deep breathing techniques to manage negative emotions. Following the game training, the transfer rates during the periods from week 1, week 2, week 3 to 6, and week 7 to 10 were 76% (n=19), 80% (n=20), 64% (n=16), and 56% (n=14), respectively. (b) In week 2, 45% of the 20 participants used multiple techniques. From week 3 to 6, 43.75% (n=7) of the participants continued using multiple techniques, and by week 7 to 10, 42.8% (n=6) of the 14 participants who maintained transfer still used multiple techniques.

skills. The interview questions particularly focused on four topics: what factors facilitated the transfer, the enhancement of skills through game training, the details of technique transfer including its emergence and gradual decline during the 10-week experiment, and the contexts in which breathing techniques were applied.

6.2.1 Improvement of breathing abilities helps transfer.

In the week 2 (14-day) follow-up interviews, 84.6% (n=22) of participants reported that changes in their breathing abilities helped them transfer the breathing techniques as ER strategies.

(1) **Improve awareness of their breathing:** 72% (n=18) participants reported a significant increase in their awareness of their breathing. P23 remarked, "After using the game, I am now more frequently aware of my breathing than before, when I face the challenge I will try to use breathing techniques I learned from the game."
(2) **Improve abilities to use breathing techniques**: 64% (n=16) of participants indicated that they had acquired additional breathing techniques after using ***Breath of Life***. Six participants reported improved control over their breathing rhythm and frequency. Notably, participants 18 and 13 mentioned that they paid more attention to the timing and ratio of inhalation and exhalation.
(3) **Longer breaths:** 32% (n=8) participants described experiencing longer breaths, P1 noted, *"Before the game training, I felt breathless and short of breath. After a few days of training, I am now able to perform deep abdominal breathing."*

6.2.2 Positive emotional improvement experiences help transfer.

During post-game interviews, participants frequently used terms such as *calm* (n=4), *relaxed* (n=12), *focused* (n=13), and *pleasant* (n=9), to describe their feelings during the game, indicating that these emotions were commonly experienced throughout the gameplay. 10 participants mentioned that they experienced the benefits of using breathing techniques during the game, which motivated them to apply these techniques in their real lives. P12 stated, "After participating in several sessions of training when I was particularly angry in real life, I instinctively started using the breathing techniques."

6.2.3 The transfer of breathing techniques evolves and diminishes over time.

During interviews on week 1 and week 2, Participant 9 noted, *"Starting from the third day of the game test, I began using the breathing techniques from the game before bedtime."* Participant 20 stated, *"After four days of using the game, I began using the 5:5 paced breathing technique to manage my anxiety."* Participants (n=5) reported a decline in the effectiveness of the game training. Participant 13 reported, *"During the 7 days of using the game, the long breathing techniques for emotion regulation gradually integrated into my daily life. However, by the third day after discontinuing the game, my awareness of breathing began to wane."* Participant 9 stated, *"After the game-based training ended, I used the breathing techniques intermittently in daily life, but not as frequently as during the testing period of the game training. I noticed that, over time, I gradually forgot how it felt to use the breathing techniques during the game training."*

6.2.4 Diversity of contexts in which breathing techniques are utilized.

Most participants primarily used these techniques to address negative emotions and before bedtime. By week 2, 47.36% (n=9) of the 19 participants reported using breathing techniques when **experiencing tension and anxiety,** while 50% (n=8) applied them in **stressful** situations. Some participants also reported they will use the breath technique when they feel **bored, fatigued, or have significant emotional fluctuations**. P23 noted, *"When my boyfriend annoys me, I use breathing techniques to manage my emotions and avoid arguments."* 7 participants mentioned using breathing techniques to assist with sleep.

7 DISCUSSION AND FUTURE WORK

Personalized frequency and duration of usage among users. As shown in section 6.1, while most participants were able to apply the breathing techniques flexibly in the weeks following the training, the frequency of use gradually declined over time. Section 6.2.3 further highlights that some participants forgot to use the techniques or experienced a decline in their ability to apply them effectively. To address these challenges, future designs could incorporate personalized adaptation strategies that account for individual differences in usage frequency and duration. Such strategies could help sustain engagement and improve the long-term effectiveness of breathing techniques in real-world contexts.

Timely reminders help promote their skill transfer. According to the results of section 6.2.1, It was surprising to observe an increase in the transfer rate of breathing from 7-14 days (week 2). In interviews in week 2, 8 participants mentioned that daily self-reports enhanced their awareness of breathing techniques. Therefore, future designs could incorporate timely reminders to help promote their skill transfer.

Long-term game engagement and transfer evaluation. At the end of the experiment, 72% of the participants expressed interest in accessing a web version of *Breath of Life* for daily self-use, reflecting high user acceptance and potential application value. However, This study has two main limitations. First, it relies on self-reported measures, which may introduce biases such as recall bias or social desirability bias. Additionally, the study did not incorporate physiological signal monitoring, which could have provided a more granular understanding of the long-term effects of the breathing techniques. Second, the sample size was relatively small and primarily consisted of healthy female university students, potentially restricting the generalizability of the findings to broader populations, including individuals of different ages, genders, and mental health conditions. In the next phase, we plan to refine the game design, expand the sample to include individuals with emotional disorders and diverse age groups while ensuring gender balance, introduce a control group, and integrate wearable devices for real-time physiological monitoring to comprehensively evaluate game engagement and skill transfer effects.

8 CONCLUSION

We designed a breathing training game to help users transfer breathing skills into real-life contexts. After one week of use and nine weeks of testing, results confirmed that *Breath of Life* promotes skill transfer. Cross-sectional comparisons showed that 84% of the participants achieved transfer to real life during the 10 weeks, with 71.4% of participants using 2-4 techniques. This further validated the effectiveness of the design considerations and provided insights for fostering long-term transfer.

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A Questionnaires

A.1 Questionnaire 1:Pre-Test Breathing Techniques Usage Habits

Q1 How do you typically regulate your emotions when experiencing negative emotions? [Fill in the blank with your answer]
Q2 Do you use breathing techniques to manage your emotions when experiencing negative emotions?
(Single choice) ( ) Yes ( ) No
Q3 If yes, how do you use breathing to regulate your emotions? [Fill in the blank with your answer]

A.2 Questionnaire 2: Transfer and Application of Breathing Techniques in Daily Life (week 1)

Q1 Did you apply the breathing techniques learned from the 'Breath of Life' game in your daily life?[Single choice]
( ) Every day ( ) 5-6 days ( ) 4-5 days ( ) 2-3 days ( ) 1 day ( ) Never
Q2 Which breathing techniques have you used In the past 4 weeks? [Multiple choice]( ) Long breathing ( ) 5-5 breathing ( ) Tactical breathing ( ) 4-7-8 breathing( ) Other [Fill in the blank with your answer]

A.3 Questionnaire 3: Daily Self-Reports (Week 2, From Days 7 to 14)

Q1 Date: [Fill in the blank with your answer]
Q2 Did you use specific breathing techniques today? [Single choice]
( ) Yes ( ) No
Q3 If you used specific breathing techniques today, please describe: in what situations did you use them? Which specific techniques did you use? [Fill in the blank with your answer]
Q4 How did you feel when you used the breath technique today? [Fill in the blank with your answer]
Q5 Why did you use breathing techniques today? [Fill in the blank with your answer]

A.4 Questionnaire 4: Transfer and Application of Breathing Techniques for week 3-6, week 7-10

Q1 In the past 4 weeks, have you consciously or unconsciously used the breathing techniques you learned in the game? [Single choice] ( ) Yes ( ) No
Q2 If you have not used any breathing techniques, what was the reason? [Fill in the blank with your answer]
Q3 In what situations did you use breathing control or specific breathing methods? [Fill in the blank with your answer]
Q4 How frequently have you used the breathing techniques? [Multiple choice]( ) Every day( ) 5-6 days per week ( ) 3-4 days per week ( ) 1-2 days per week( ) A few times per month( ) Only 1-2 times in specific situations [OR Fill in the blank with your answer]
Q5 Which breathing techniques have you used In the past 4 weeks? [Multiple choice]( ) Long breathing ( ) 5-5 breathing ( ) Tactical breathing ( ) 4-7-8 breathing( ) Other [Fill in the blank with your answer]

B Semi-Structured Interview Outline

B.1 Semi-Structured Interview Outline 1 after every training

Q1 How did you feel physically during today's breathing exercises? (Ask daily)
Q2 Which breathing technique did you prefer during today's training? Which technique was most challenging for you, and why? (Ask daily)
Q3 Did you consciously adjust your breathing during today's training? If so, please describe the circumstances under which you made these adjustments and the specific methods you used. (Ask daily)
Q4 How would you evaluate your overall game experience today? What difficulties did you encounter? (Ask daily)
Q5 How does today's game experience differ from your previous experiences? (Ask daily from the second day)
Q6 During the test period, have you tried applying breathing regulation techniques in your daily life? If so, please specify which techniques you used and the contexts in which you applied them. (Ask daily from the second day)
Q7 On approximately which day did you begin to notice a significant difference in your experience compared to the earlier days? (Ask on the seventh day of training)

B.2 Semi-Structured Interview Outline 2

Q1 After the whole training, have you mastered any breathing techniques? Which techniques have you learned?
Q2 Do you think that using breathing techniques in daily life is effective for emotional regulation?
Q3 What differences do you perceive between using breathing techniques in the game and applying them in your daily life?
Q4 Have you continued to use the breathing techniques in your daily life? If so, please describe the specific contexts in which you used them and why?

B.3 Semi-Structured Interview Outline 3

Q1 Did you use different breathing techniques during past 5 weeks/ 9 Weeks? If so, please explain the reasons for your choices.
Q2 Why do you think you continued to use breathing techniques in daily life one month after playing the game?
Q3 Please describe in detail the scenarios in which you used breathing techniques